

Class 17: Rank and Dimension

Definition: The dimension of a nonzero subspace H is

The dimension of the zero subspace is defined to be

Examples: The dimension of $\mathbf{R}^n =$

Let A be any $m \times n$ matrix with m rows and n columns.

The dimension of the column space of A is

The dimension of the null space of A is

Definition: The rank of a matrix A is

Definition: The nullity of a matrix A is

The Rank Theorem: If A has n columns then the rank of A + the nullity of A is equal to n .

Proof:

Some "Think about it" questions

19. If the subspace of all solutions of $A\mathbf{x} = \mathbf{0}$ has a basis consisting of three vectors and if A is a 5×7 matrix, what is the rank of A ?

20. What is the rank of a 6×8 matrix whose null space is three-dimensional?

24. Construct a 3×4 matrix with rank 1.

Coordinate Systems

The standard coordinates for the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ are

Now if $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is any basis for $\mathbf{R}^n$ and $\mathbf{x}$ is any vector in $\mathbf{R}^n$ we define the coordinates for $\mathbf{x}$ relative to basis B to be the unique weights used to express $\mathbf{x}$ as a linear combination of the basis vectors.

The coordinate vector for $\mathbf{x}$ relative to the basis B is denoted by

Example: Let $\vec{b}_1 = \begin{bmatrix} -3 \\ 2 \\ -4 \end{bmatrix}$ and $\vec{b}_2 = \begin{bmatrix} 7 \\ -3 \\ 5 \end{bmatrix}$ and $\vec{b}_3 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

What is the coordinate vector for $\vec{x} = \begin{bmatrix} -1 \\ 6 \\ 2 \end{bmatrix}$
relative to the basis $B = \{ \vec{b}_1, \vec{b}_2, \vec{b}_3 \}$